

Problem 1: Use cards A–H to make a reversible key: each letter has one output, and each output is used once. Build the four sets of loops below. Move a counter one arrow per turn on each loop. Record the first turn when all counters are home.

Loop sizes	My key or arrow loops	Return
8		
4 + 4		
6 + 2		
5 + 3		

Problem 2: For the 5+3 key, list the return turns of each separate loop through turn 20. Circle the first turn on both lists. Repeat for 4+4. Compare with one loop of size 8.

Problem 3: Find the slowest eight-letter key. Sort possible loop sizes by the largest loop. Use eight cards to check that your sizes total eight. For largest loops 4–8, list the splits of the leftover cards and find the best return. In the last row, use loop sizes 1,2,3 to find one turn that returns every loop. A fixed letter is a loop of size 1.

Largest loop	Possible leftover loops	Best return
8	none	
7		
6		
5		
4		
At most 3		

Problem 4: Compare the best return in every row with your page-1 key. Record the winning loop sizes, key, and return time.

Problem 5: Add card I. Build keys with loops $5+4$, $7+2$, and $4+3+2$. List each loop's return turns until they first agree. Record the whole-key return for each.

Problem 6: Classify nine-letter keys by largest loop, as on page 2. Find the best return for largest loops 9,8,7,6,5,4. When all loops have size at most 3, give one turn that returns all of them. Compare these bounds with your constructions.

Problem 7: Use six cards. Test loops $3+2+1$, 6, and $5+1$. Compare with the five-card $3+2$ key. Predict whether adding a card must always make the longest possible return strictly larger. Record a supporting example and a question still to check.